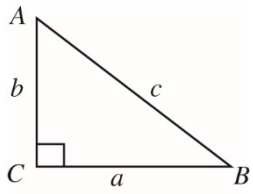
Solve a Right Triangle

Pythagorean theorem: $a^2 + b^2 = c^2$ $\quad A + B + C = 180°$

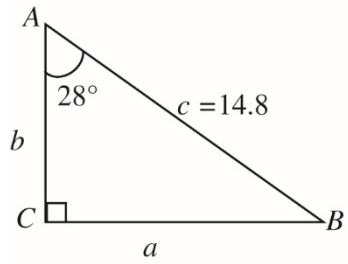


$$\sin\theta = \frac{\text{opp}}{\text{hyp}},\ \cos\theta = \frac{\text{adj}}{\text{hyp}},\ \tan\theta = \frac{\text{opp}}{\text{adj}}$$

$$\csc\theta = \frac{\text{hyp}}{\text{opp}},\ \sec\theta = \frac{\text{hyp}}{\text{adj}},\ \cot\theta = \frac{\text{adj}}{\text{opp}}$$

---

1. Solve the right triangle. Round the lengths of sides to 1 decimal place.



$B = 180 - 28 - 90$

$B = 62$

$a^2 + 13.1^2 = 14.8^2$

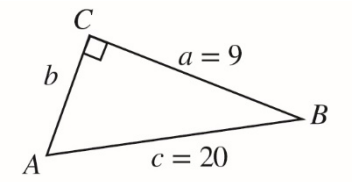
$a = 6.9$

$\sin 62 = \frac{b}{14.8}$

$b = 14.8 \sin 62$

$b = 13.1$

2. Solve the right triangle. Round the measurements to 1 decimal place.



$B = 180 - 90 - 26.7$

$9^2 + b^2 = 20^2$

$b^2 = 319$

$b = \sqrt{319}$

$b = 17.9$

$\sin A = \frac{9}{20}$

$A = \sin^{-1}\left(\frac{9}{20}\right)$

$A = 26.7$

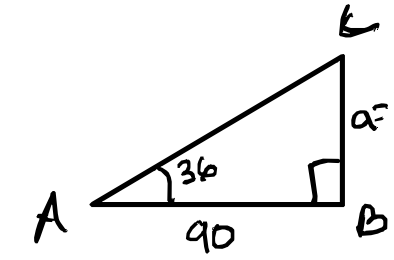
$B = 63.3$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

Solve Applications of Right Triangles

3. To determine the height of a window in the top of a lighthouse, two measurements are taken from the ground 90 feet from the base of a lighthouse. The angle of elevation from the ground to the bottom of the window is 36° and the angle of elevation from the ground to the top of the window is 42°.

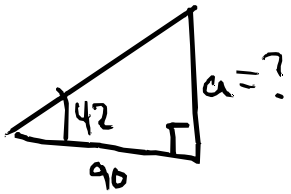
   Find the height of the window to the nearest foot.



$\tan 36 = \frac{a}{90}$

$a = 90 \tan 36$

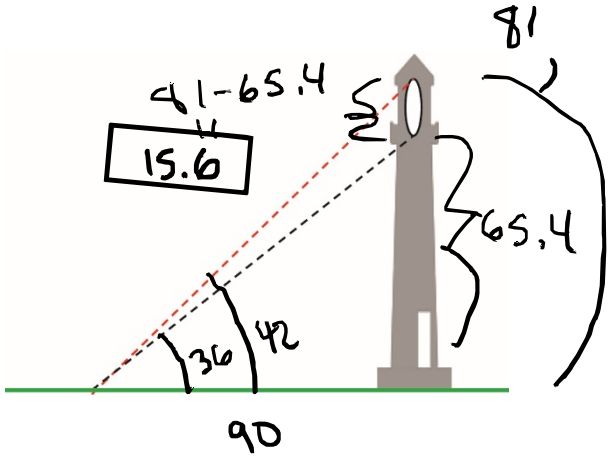
$a = 65.4$



$\tan 42 = \frac{a}{90}$

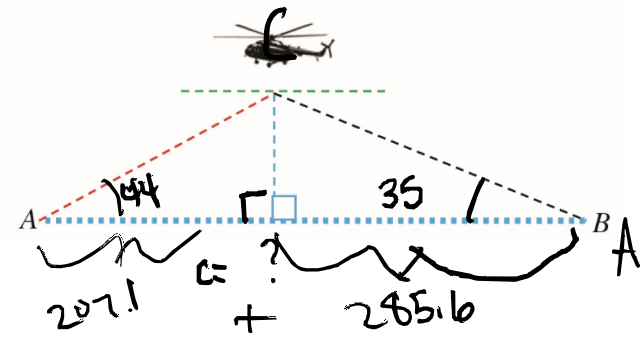
$a = 90 \tan 42$

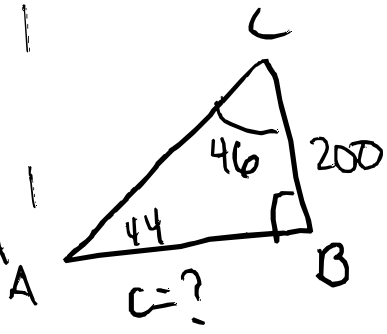
$a = 81$



   From the given information, can we determine the height of the lighthouse?

4. An observer in a helicopter hovering 200 feet above the sea makes measurements to two buoys at points *A* and *B* below. The angles of depression are 44° and 35° respectively. Assuming the observer is in the same vertical plane as the buoys, what is the distance between the two buoys? Round to the nearest foot.



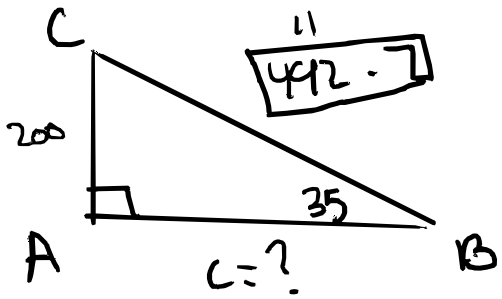


$\tan 44 = \frac{200}{c}$

$c \tan 44 = 200$

$c = \frac{200}{\tan 44}$

$c = 207.1$



$\tan 35 = \frac{200}{c}$ → $c \tan 35 = 200$

$c = \frac{200}{\tan 35}$ $c = 285.6$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

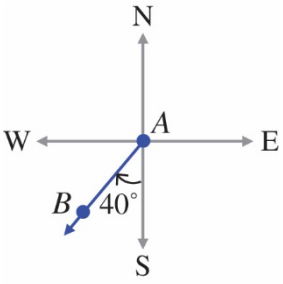
Compute the Bearing of an Object

The term *bearing* is used to convey the angular relationship from one point to another.

The **bearing** from point A to point B is the degree measure of the acute angle formed by ray $\overrightarrow{AB}$ and the north-south line.

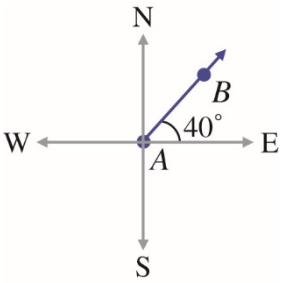
A bearing is denoted by the letter N or S (north or south), followed by the measure of the angle in degrees, followed by the letter E or W (east or west).

For example, a ship bearing S40°W is heading in a southwesterly direction. Specifically, the ship is traveling 40° to the west of due south.



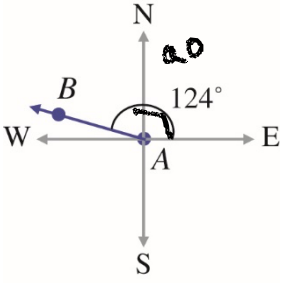
Find the bearing from A to B.

5.



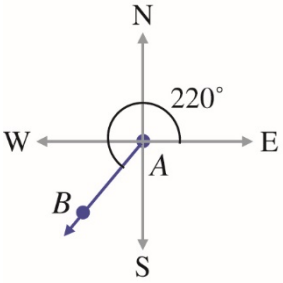
N50E

6.



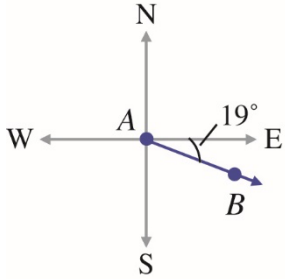
N34W

7.



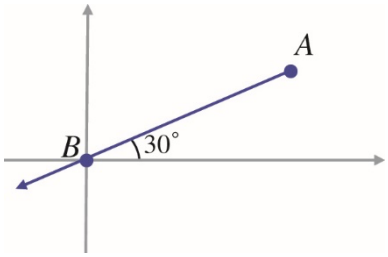
S50W

8.



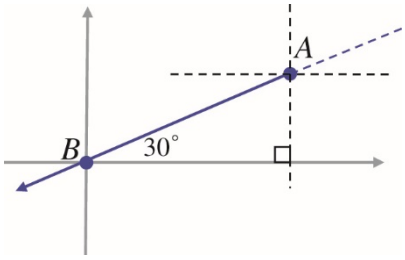
S71E

9. Find the bearing from A to B.



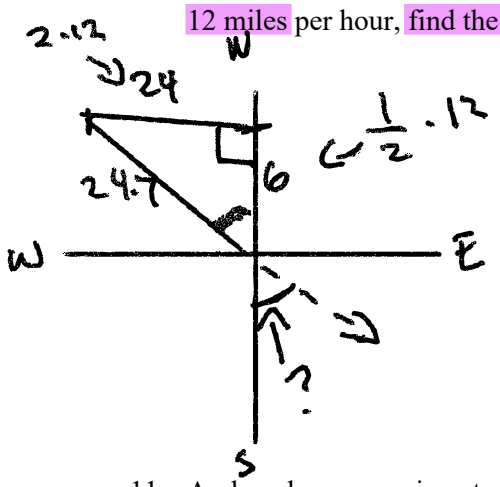
S60W

Hint: Sketch a coordinate system with origin at point A.



© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

10. A boat leaves a marina and sails north for 30 minutes, then west for 2 hours. If the boat's average speed is 12 miles per hour, find the bearing for the return trip. Round to the nearest degree.



$24^2 + 6^2 = c^2$

$c^2 = 612$

$c = 24.7$

don't need

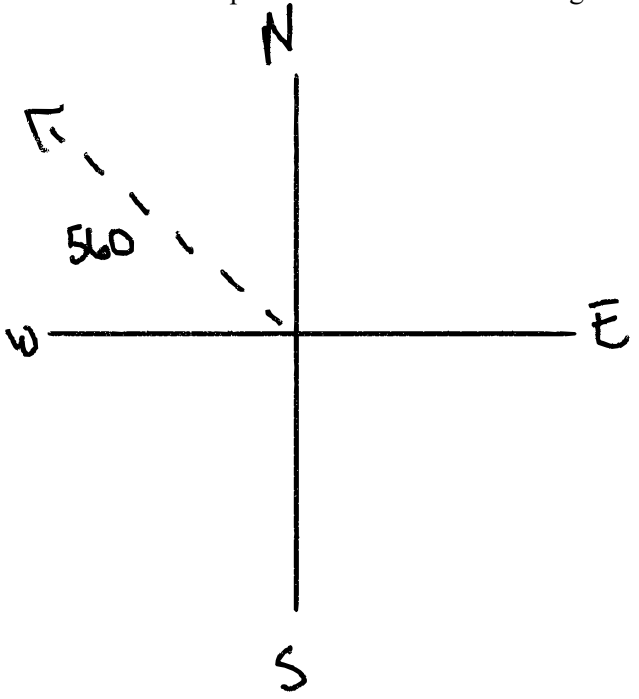
$\tan A = \frac{24}{6}$

$A = \tan^{-1}\left(\frac{24}{6}\right)$

$A = 76^\circ$

S 76 E

11. A plane leaves an airport at a bearing of N38°W and travels at 280 mph for 2 hours. The plane then makes a 90° left turn and travels 500 miles to its destination. Find the bearing required to locate the plane from the airport. Round to the nearest degree.



$280 \cdot 2 = 560$

not needed ofajk

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.